

Performance Testing

Field	Details
Date	30 September 2026
Team ID	SWTID-2026-6777
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. This document records the testing approach and available results for ComicCraft.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (API request/response testing)
Type of Testing	Load-oriented API testing; stress testing after baseline validation.
Target Module / API	/ (home page), /generate (comic generation)
Test Environment	Local FastAPI/Uvicorn environment (127.0.0.1:8000)
Test Date	30 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Home page load	Planned: 1	Planned	Page loads without server error.
2	Comic generation endpoint (/generate)	Planned: 1	Planned	Valid story idea returns a 5-frame comic without server error.
3	Generation with a long story idea (up to 600 characters)	Planned: 1	Planned	Comic returns without request failure.
4	Different story mood and art direction options	Planned: 1	Planned	Comic returns successfully for each selection.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Not measured	Pending execution	No measured load result supplied.
2	Response Time (Max)	< 5 seconds	Not measured	Pending execution	No measured load result supplied.
3	Throughput (Req/sec)	Baseline	Not measured	Pending execution	Requires concurrent requests.
4	Error Rate	< 1%	Not measured	Pending execution	Requires load-test run.
5	CPU Utilization	< 80%	Not measured	Pending execution	Requires monitoring during load.
6	Memory Utilization	< 80%	Not measured	Pending execution	Requires monitoring during load.

Step 4: Observations & Analysis

Key Findings:

ComicCraft exposes FastAPI endpoints for the home page and comic generation (/generate).

In the local demo mode, a 5-frame comic was generated and displayed in the browser; live Gemini generation requires API keys.

Baseline performance values must be collected by repeated requests.

Bottlenecks Identified:

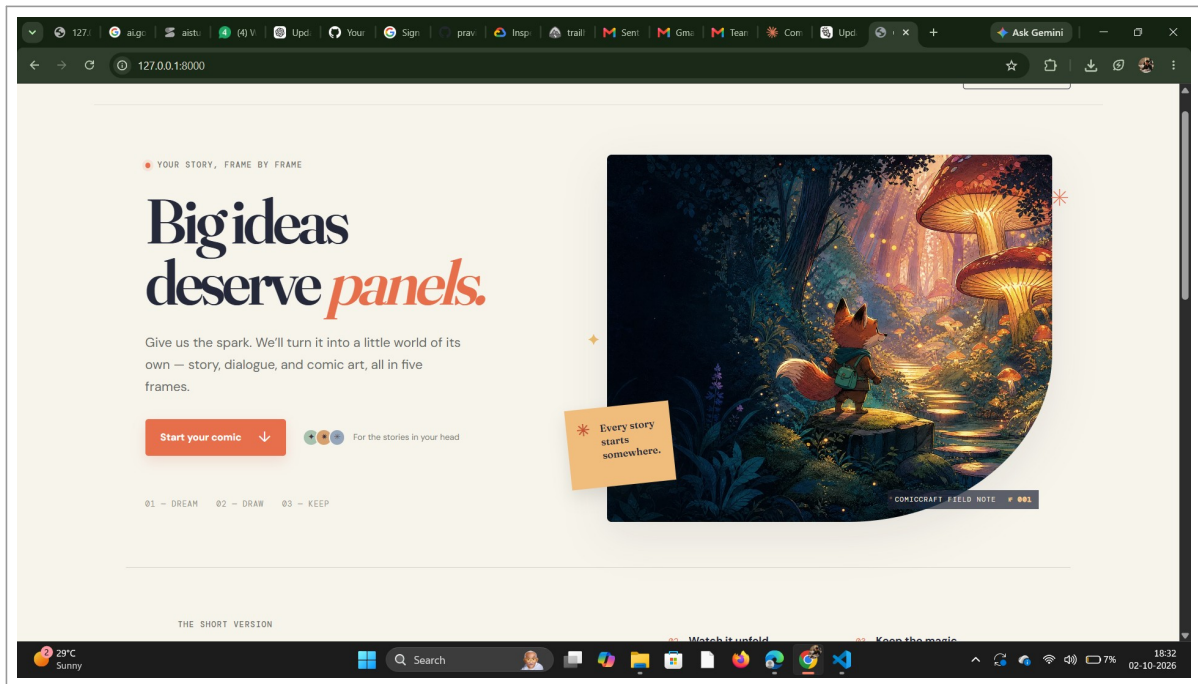
No measured bottleneck is claimed without an executed performance test.

Optimization Steps Taken:

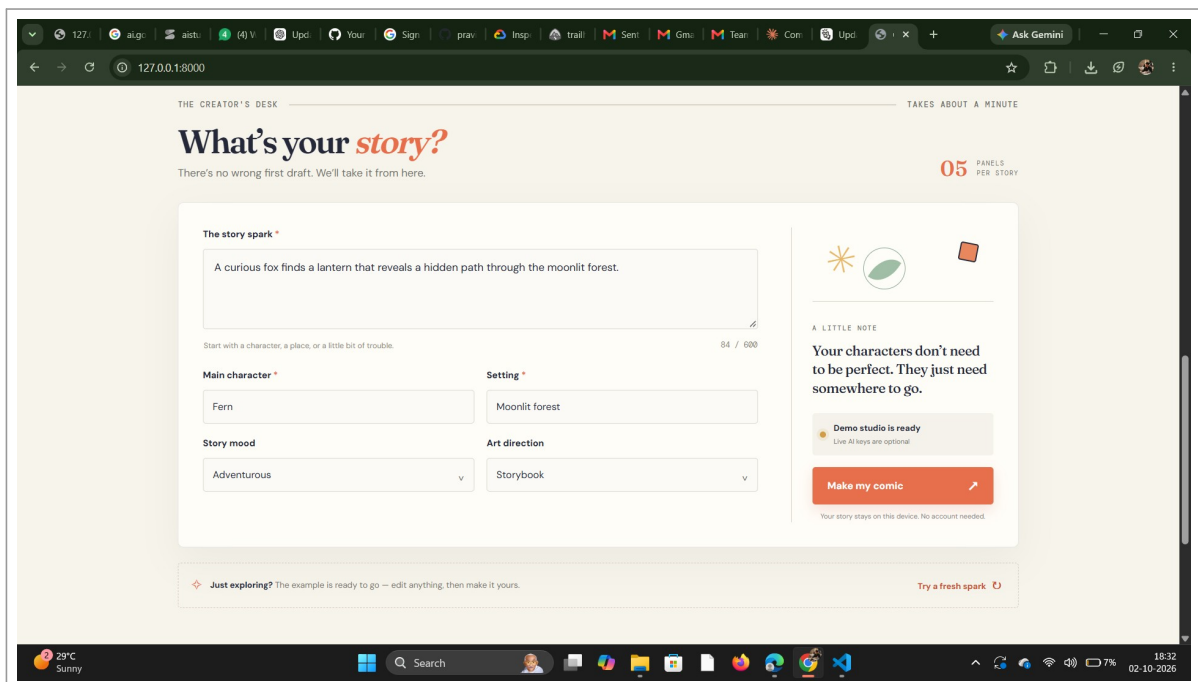
Comic generation is handled by task-specific modules, and a local demo mode allows testing without live API calls.

Step 5: Screenshots / Evidence

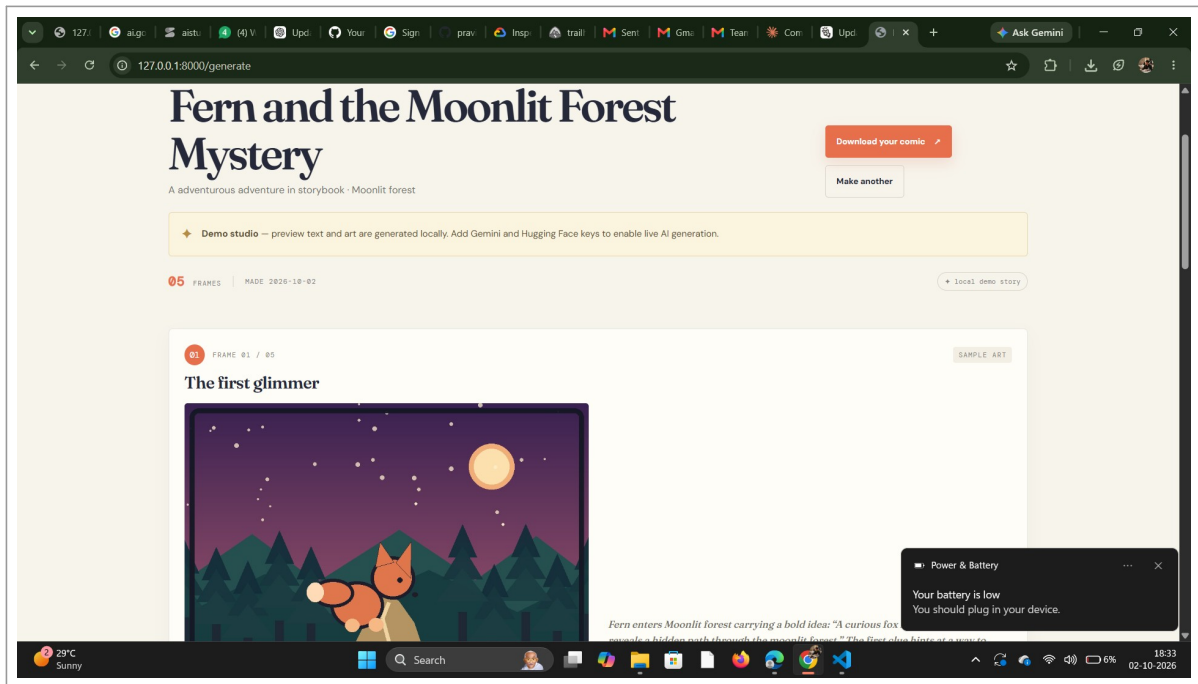
The screenshots below show the ComicCraft application running on the local FastAPI/Uvicorn environment. No load-test tool results are included because no load test has been executed yet.



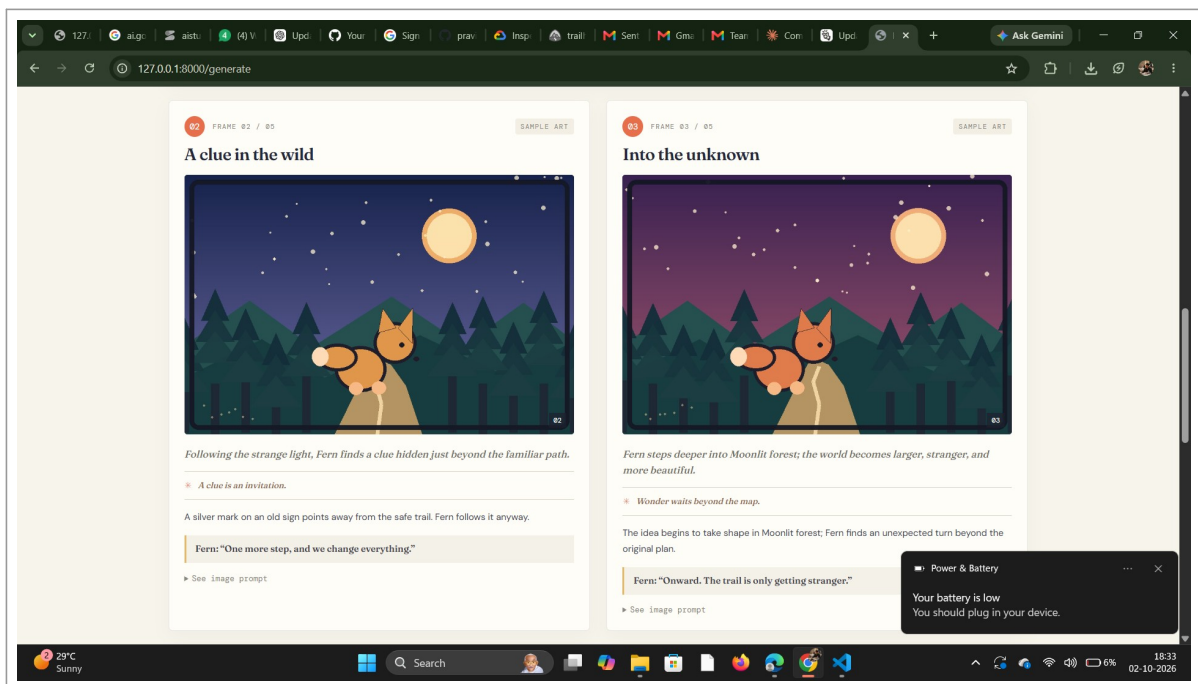
Evidence 1 - ComicCraft landing page ("Big ideas deserve panels.") running locally at 127.0.0.1:8000.



Evidence 2 - Story input form with story spark, main character, setting, story mood, art direction, and the "Make my comic" button.



Evidence 3 - Generated comic result at /generate: "Fern and the Moonlit Forest Mystery" (5 frames) with download and "Make another" options.



Evidence 4 - Generated comic frames 2 and 3 showing scene art, narration, and character dialogue.